EXPERIMENT NO. 2

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Input Code:

```
#include <stdio.h>
int binarySearch(int arr[], int left, int right, int target) {
    if (right >= left) {
        int mid = left + (right - left) / 2;
        if (arr[mid] == target)
            return mid;
        if (arr[mid] > target)
            return binarySearch(arr, left, mid - 1, target);
        return binarySearch(arr, mid + 1, right, target);
    }
    return -1;
}
int main() {
    int arr[] = {2, 3, 4, 10, 40};
    int n = sizeof(arr) / sizeof(arr[0]);
    int target = 10;
    int result = binarySearch(arr, 0, n - 1, target);
    if (result == -1)
        printf("Element is not present in array\n");
    else
        printf("Element is present at index %d\n", result);
    return 0;
}
```

Output:

```
Element is present at index 3
```